

Anodic Stripping voltammetry

* This technique is used to identify the metal ions in the solution.

Steps

(A) Pre-concentration step

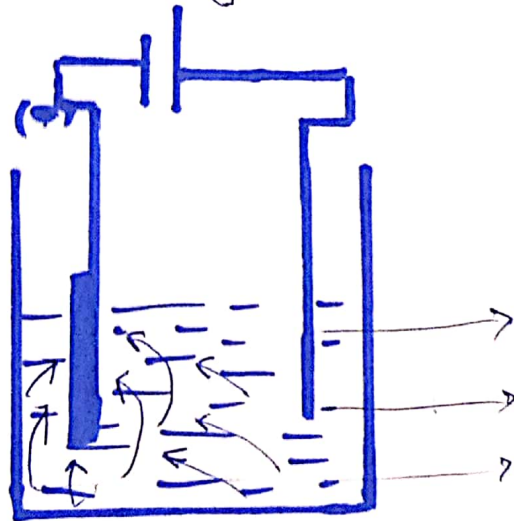
In this step all the metal ions present in the solution are deposited on the surface of working electrode.

working electrodes

→ Hanging mercury drop electrode, Gold and Pt electrode

2) In this step we select the suitable range of potential which can reduce all the metal ions present in the solution.

Suppose a solution containing Cu^{2+} , Cd^{2+} and Pb^{2+} ions with their reduction ranges **20V, 40V, 60V** respectively.



Cu^{2+}	(20V)
Cd^{2+}	(40V)
Pb^{2+}	(60V)

→ using thin film mercury electrode.

when we apply **100V** potential through the solution then all the metals ions present in the solution reduce and deposit on the mercury electrode and forms **amalgam**,

→ which depends upon the

- a) current flowing through cell
- b) conc. of reducible ions.

B) Quiescent or rest period

In this step stirrer is switched off. results formation of uniform amalgam layer

(4)

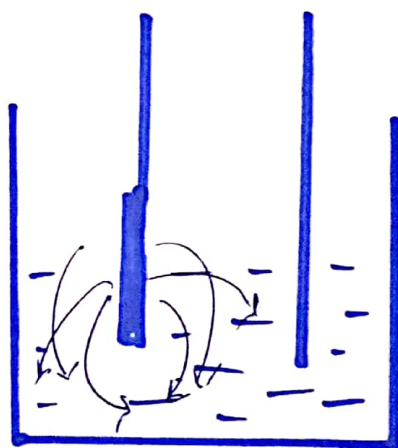
on the surface of mercury electrode, named as rest period.

→ This step is not necessary in case of solid electrode, only for mercury electrode.

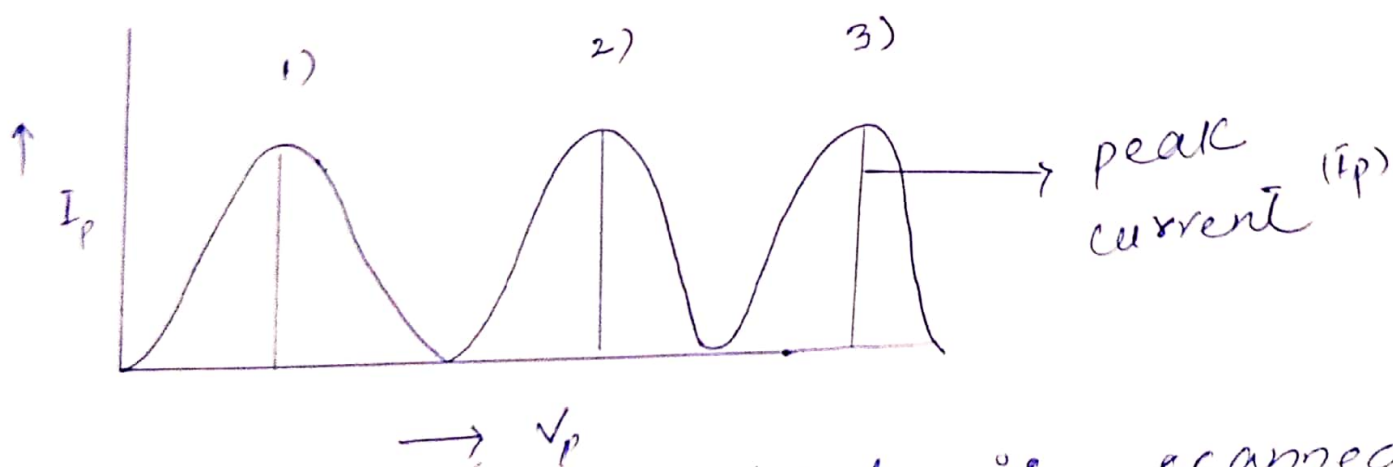
c) stripping process

In this step the applied potential is scanned in reverse direction continuously with time.

→ As we approach oxidation potential of Pb^{2+} 60V, all the Pb^{2+} ions ^{will} be stripped off



5) in the solution. As Pb^{2+} ions are stripped in the solution, the current of solution gradually increases and then decreases, obtaining a peak, known as anodic stripping voltamogram.



→ As the potential is scanned continuously, at $40V$ all the Cd^{2+} ions will be stripped out of the solution and obtained.

(b)

another peak and same at
20V all the **Cu^{2+}** ions will
stripped off and gave third
peak.

→ According to the concentration
of metal ions in the solution,
they will show relative
peak current.

→ The metal ion which is
present in greater quantity
will show greater current
peak and can be identified
by stripping off at their
respective voltage.

By : Jamal Ahmad #45